

Text 1

Weather and climate researchers have long been interested in climate change models. During recent years, the mechanisms and dynamics of atmospheric particle growth have become better understood. However, the influence of particle size on cloud formation and the radiative balance of the Earth is still not taken into climate change models. Researchers have reviewed the past 20 years' results of research on nano-sized particles in the atmosphere, known as aerosol particles. Their goals were to provide tools for more accurate modeling, bridge the gap between experimental research and modeling, and reduce the uncertainty around particle growth.

The uncertainties about particle growth are of serious concern to researchers. One of the problems is a precise experiment to determine particle growth rate at several measuring sites around the world. This applies to particles with a size less than 5 nanometers.

Results from recent experiments give more knowledge on the formation and growth of various particles. However, they cannot be applied effectively in climate models because their precise model requires too much computing power.

Most existing weather forecasting models do not factor the presence of aerosols in detail. In fact, the interaction between aerosols and radiation, as well as between aerosols and clouds, affects the Earth's surface temperature. This is because aerosols affect the brightness of clouds and solar radiation reflected back by the Earth.

(Adapted from <https://www.phys.org>)

Text 2

Aerosols are liquid droplets or fine solid particles suspended in air or another gas. Depending on their kind and color, these tiny airborne particles have the ability to either cool or warm the climate. They affect the climate in two ways: by directly changing the amount of sunlight that reaches Earth's surface, and by forming clouds. Clouds are composed of small droplets, formed with the condensation of water on the surface of aerosol particles in the air.

The properties of the cloud cover are determined by the amount and composition of aerosol particles in the cloud. If the amount of water in the air is constant, a large quantity of aerosols will result in a small droplet size but a large surface area. This will result in the formation of bright clouds that scatter sunlight more effectively.

Gases and other emissions caused by human activities also generate new particles in the atmosphere. These particles could grow large enough to affect the formation of clouds and, ultimately, the climate.

The growth of aerosol particles is a sensitive process affected by atmospheric conditions and the gases in the air. The formation and growth of new particles in the atmosphere can be observed on clear and sunny days. Several types of molecules form clusters that can become condensation sites for other molecules, resulting in larger particles that eventually form clouds.

Questions 15 to 20

1. According to Text 1, what is the main problem with the uncertainty of particle growth in clouds?

A. There are too many types and properties of aerosol particles contributing to the growth of other various particles in the clouds

B. The growth process cannot be applied in climate models because they require complex mathematical calculations

C. The formation and growth of particles in the atmosphere can be observed only on clear and sunny days

D. The growth is a sensitive process affected by atmospheric conditions and the gases in the air

E. Precise observations to determine particle growth rate at different places are still absent

2. The word "they" in Text 2, paragraph 1, refers to...

A. Particles

B. Aerosols

C. Droplets

D. Clouds

E. Ways

3. Which of the following sentences from Text 1 is an opinion?

A. Weather and climate researchers have long been interested in climate change models

B. This applies to particles with a size less than 5 nanometers

C. Results from recent experiments give more knowledge on the formation and growth of various particles

- D. Most existing weather forecasting models do not factor the presence of aerosols in detail
- E. This is because aerosols affect the brightness of clouds and solar radiation reflected back by the Earth

4. What is the relationship between Texts 1 and 2?

- A. Both texts illustrate the physical and chemical properties of aerosol particles in the clouds and their relationship with global warming
- B. Both texts mention the difficulties of weather forecasting and research development about cloud formation
- C. Text 2 discusses the formation of nano-sized atmospheric particles, and Text 1 discusses their growth
- D. Text 1 describes the influence of the size of atmospheric particles, whose characteristics are explained in Text 2
- E. Text 1 explains the development of a weather forecasting model by considering the influence of atmospheric particles described in Text 2

5. The information on how aerosols in the atmosphere affect the Earth's surface can be found in...

- A. Text 1, paragraph 1; Text 2, paragraph 1
- B. Text 1, paragraph 2; Text 2, paragraph 2
- C. Text 1, paragraph 2; Text 2, paragraph 3
- D. Text 1, paragraph 4; Text 2, paragraph 1
- E. Text 1, paragraph 4; Text 2, paragraph 4

6. Based on the information from the two texts, which of the following will most likely happen in the future?

- A. The formation and growth of atmospheric particles will be controlled by developing more accurate weather predictions
- B. The properties of the cloud cover will be easily determined by counting the amount and the composition of aerosol particles in the cloud
- C. A better weather forecasting model will be developed by including the existence of atmospheric particles to predict climate change
- D. Weather and climate researchers will be able to understand the mechanism of sunlight reflection into space due to aerosol formation in the cloud

E. The experiment on the formation and growth of various atmospheric particles will be replaced by less power-consuming simulation and modelling

PEMBAHASAN

1. Apa masalah utama terkait ketidakpastian pertumbuhan partikel dalam awan menurut Text 1?

Text 1 bilang: masalah besarnya adalah eksperimen presisi di berbagai tempat dunia itu masih kurang untuk ngukur laju pertumbuhan partikel kecil banget (<5 nanometer).

Jawaban: E. Precise observations to determine particle growth rate at different places are still absent

2. Kata "they" di Text 2, paragraf 1, merujuk ke apa?

Paragraf 1:

"Depending on their kind and color, these tiny airborne particles have the ability to either cool or warm the climate. They affect the climate..."

"These tiny airborne particles" = aerosols.

Jawaban: B. Aerosols

3. Kalimat mana dari Text 1 yang merupakan opini?

Yang bentuknya pendapat atau penilaian subjektif biasanya opini.

Kalimat A:

"Weather and climate researchers have long been interested in climate change models."

Ini lebih ke pendapat umum, bukan fakta terukur.

Jawaban: A. Weather and climate researchers have long been interested in climate change models

4. Apa hubungan antara Text 1 dan Text 2?

Text 1 → Tentang pentingnya pertumbuhan partikel dan pengaruhnya di model iklim.

Text 2 → Menjelaskan karakteristik aerosol (partikel kecil) dan bagaimana mereka mempengaruhi cuaca/iklim.

Jadi hubungan mereka:

Text 1 ngomongin pengaruh ukuran partikel, sedangkan karakteristik partikel dijelaskan di Text 2.

Jawaban: D. Text 1 describes the influence of the size of atmospheric particles, whose characteristics are explained in Text 2

5. Informasi tentang bagaimana aerosol mempengaruhi permukaan Bumi ada di mana?

Text 1, paragraf 4:

"...interaction between aerosols and radiation... affects the Earth's surface temperature."

Text 2, paragraf 1:

"They affect the climate... by directly changing the amount of sunlight that reaches Earth's surface."

Jawaban: D. Text 1, paragraph 4; Text 2, paragraph 1

6. Berdasarkan kedua teks, apa yang paling mungkin terjadi di masa depan?

Karena sekarang model prediksi iklim masih kurang akurat karena belum mempertimbangkan aerosol...

Maka logis kalau ke depan: model prakiraan cuaca akan diperbaiki dengan mempertimbangkan partikel atmosfer ini.

Jawaban: C. A better weather forecasting model will be developed by including the existence of atmospheric particles to predict climate change